先进计算与数字工程研究中心

PATTERN RECOGNITION AND MACHINE LEARNING

Lecture 04 – Discriminative Classifiers I: Linear, Generalized, Piecewise, and Fisher

Chizhi Chris ZHANG

zhangchizhi@ciomp.ac.cn

Advanced Computing and Digital Technology Research Center

University of Chinese Academy of Sciences

Fall 2026

Three periods: geometry, projection, richer representations

Time	What we connect	What you will calculate
0–45 min	Score → line → distance → multiclass rule	Labels, boundary, and signed distances
45–90 min	Labelled points → scatter → Fisher direction	One complete projection and threshold
90–135 min	Feature maps/local pieces → nonlinear regions	XOR, interval, and prototype decisions

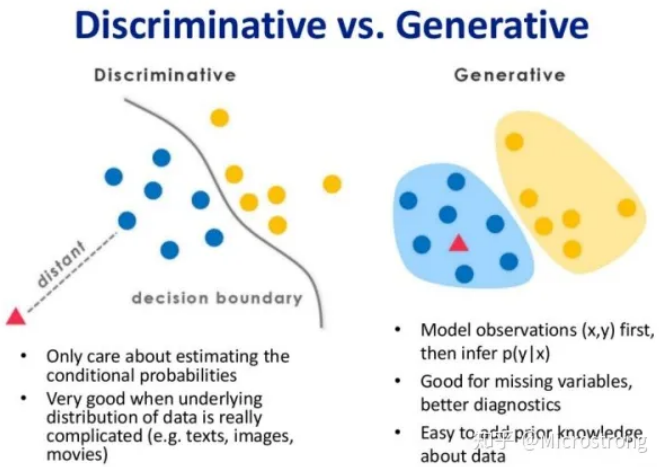
Continuity. Shared-covariance Gaussian classification already produced a linear score. We now study its geometry and learn a direction from class statistics. Perceptron mistake updates belong to the next lecture; full logistic fitting appears in the linear-model lecture. Here we build the meaning of their decision functions.

Generative and discriminative models target different objects

Route	What is fitted?	What can follow?
Generative	$p(x \mid y)$ and $p(y)$	Bayes posterior, class scores, plausible sample generation
Conditional discriminative	$p(y \mid x)$	Label probabilities and expected-risk decisions
Score discriminative	$g(x)$ or class scores	Ranking and decision boundaries

Tomato measurements. Color and firmness can be modelled by class distributions or used directly in a pack/hold score. Fitting the complete measurement distribution can help with credible structure and few labels. Fitting the boundary can avoid irrelevant distribution detail. Neither name guarantees better prediction; the assumptions and required output matter. **Probability-model preview.** Logistic regression specifies $P(1 \mid x) = 1/(1 + e^{-g(x)})$; its fitting objective is taught later. A link must be specified before a raw score is read as a probability.

Generative and discriminative models target different objects



A score needs an output interpretation

The same sorting rule can be written as $g = x_1 - .8x_2 - 3.5$ or $10g$.

$$g(7.2, 4.1) = .42, \qquad 10g(7.2, 4.1) = 4.2.$$

Both scores choose the positive class, but their magnitudes differ. If a logistic link were specified, the corresponding numbers would give different probabilities:

$$\sigma(.42) \approx .603, \qquad \sigma(4.2) \approx .985.$$

Three separate statements

A score defines an ordering. A threshold defines a decision. A probability model defines how numerical scores describe uncertainty.

The link illustrates the distinction; its parameters must be fitted and assessed before these values support probability claims.

Bayes can yield the same linear decision form

For two Gaussian classes sharing Σ, posterior comparison yields

$$g(x) = \log \frac{p(x \mid 1)\pi_1}{p(x \mid 2)\pi_2} = w^T x + b,$$

$$w = \Sigma^{-1}(\mu_1 - \mu_2), \quad b = -\tfrac{1}{2}\mu_1^T \Sigma^{-1}\mu_1 + \tfrac{1}{2}\mu_2^T \Sigma^{-1}\mu_2 + \log \frac{\pi_1}{\pi_2}.$$

Decision. Choose class 1 if $g(x) > 0$ under zero-one loss. **Two meanings of the same formula.** A generative model derives w, b from class statistics. A discriminative learning rule can fit w, b directly. The geometry $g(x) = 0$ is identical once parameters are fixed. Now study what the direction, intercept, and scale of this score actually do.

One score links features, boundary, and class label

Teaching model. Color x_1, firmness x_2, and score

$$g(x) = x_1 - .8x_2 - 3.5 = w^T x + b, \quad w = (1, -.8)^T.$$

Input	Score	Decision
$(7.2, 4.1)$	.42	Pack now, $+1$
$(5.4, 6.0)$	-2.90	Hold, -1
$(6.8, 3.5)$	.50	Pack now, $+1$

Boundary. $x_1 - .8x_2 - 3.5 = 0$. **Rule.** Sign turns a continuous score into a label. Specify a tie rule at zero. The magnitude is not automatically a probability; multiplying every parameter by 10 leaves all labels unchanged.

A complete classifier states its equality policy

Use $g = x_1 - .8x_2 - 3.5$ with positive labels for packing.

Input	Computation	Stated output
$(7.2, 4.1)$	$7.2 - 3.28 - 3.5 = .42$	Pack
$(7.5, 5)$	$7.5 - 4 - 3.5 = 0$	Hold under $g \leq 0$ policy
$(5.4, 6)$	$5.4 - 4.8 - 3.5 = -2.9$	Hold

Decision specification

The equation $g = 0$ locates the boundary; inequalities assign its sides. Equality must belong to a stated class or review action.

A review band such as $|g| < .2$ is an additional policy. It depends on score scaling, so it must be transformed if all parameters are rescaled.

Slope and intercept describe the same boundary

For $g(x) = w_1 x_1 + w_2 x_2 + b$, when $w_2 \neq 0$,

$$x_2 = -\frac{w_1}{w_2} x_1 - \frac{b}{w_2}.$$

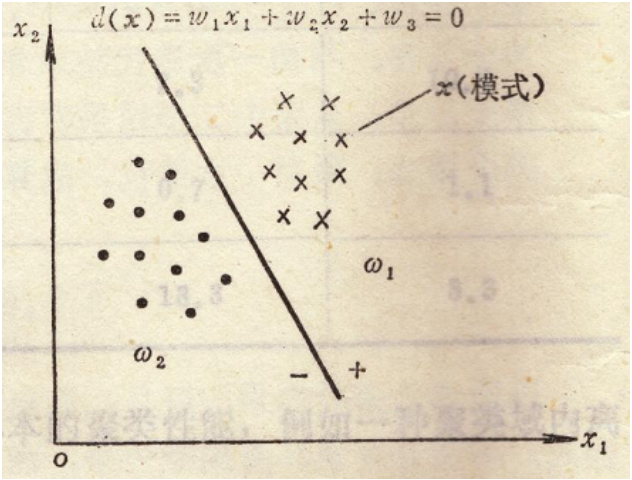
Our sorting score gives

$$x_2 = 1.25 x_1 - 4.375.$$

Boundary point	Score check	Which side packs?
$(3.5, 0)$	$3.5 - 0 - 3.5 = 0$	Larger x_1 at fixed x_2
$(7.5, 5)$	$7.5 - 4 - 3.5 = 0$	Smaller x_2 at fixed x_1

In d dimensions, $w^T x + b = 0$ is a hyperplane. If $w_2 = 0$, do not divide by zero: the boundary is vertical when $w_1 \neq 0$. A line equation must be paired with a class-side rule.

Slope and intercept describe the same boundary



Vertical boundaries and constant classifiers

Slope form requires a nonzero coefficient of the vertical coordinate. The affine form also handles the remaining cases.

Parameters	Score equation	Geometry and prediction
$w = (2, 0), b = -6$	$g = 2x_1 - 6$	Vertical line $x_1 = 3$; positive to its right
$w = (0, 2), b = -6$	$g = 2x_2 - 6$	Horizontal line $x_2 = 3$; positive above
$w = (0, 0), b = 2$	$g = 2$	Always positive; no boundary
$w = (0, 0), b = 0$	$g = 0$	Every input ties; equality policy decides

At $(4, 1)$, the first rule gives 2 and the second gives -4. A signed-distance formula requires $\|w\| > 0$; constant classifiers have no boundary normal to normalize.

The weight vector is normal to the boundary

For two boundary points x_a, x_b,

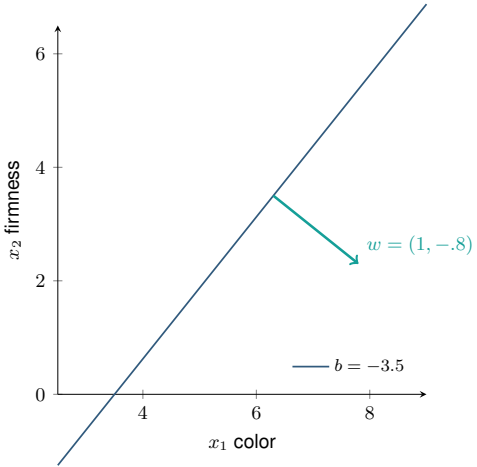
$$w^T x_a + b = w^T x_b + b = 0 \quad \Longrightarrow \quad w^T (x_a - x_b) = 0.$$

Thus w is perpendicular to every movement along the hyperplane. **Numerical check.** From $(3.5, 0)$ to $(7.5, 5)$, the tangent displacement is $(4, 5)$:

$$(1, -.8)^T (4, 5) = 4 - 4 = 0.$$

Moving a distance ϵ along the unit normal $u = w/\|w\|$ increases the score by $\epsilon\|w\|$. **Interpretation.** w specifies the direction of fastest score increase; b shifts the boundary. w lies across the boundary, not along it. This geometric role will return when Fisher selects a projection direction.

The weight vector is normal to the boundary



A bias change translates the boundary along its normal

Keep $w = (1, -.8)^T$ and change the sorting bias from -3.5 to -2.5.

$$g_{new}(x) = g_{old}(x) + 1, \qquad \Delta x = -\frac{1}{\|w\|^2} w = (-.610, .488)^T.$$

The old boundary point $(7.5, 5)$ moves to approximately $(6.890, 5.488)$:

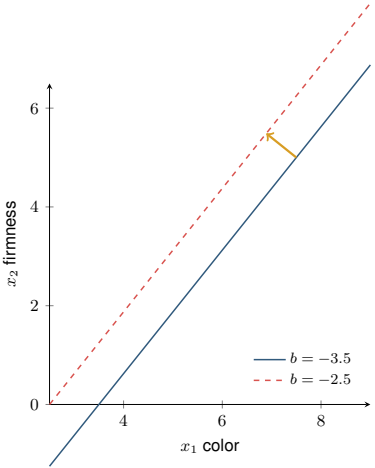
$$6.890 - .8(5.488) - 2.5 \approx 0.$$

Geometric meaning

The normal direction stays fixed. The boundary moves about .781 units toward the negative normal, and the positive half-space grows.

This is a parallel translation, rather than a rotation. For the unchanged point $(7.2, 4.1)$, the score increases from .42 to 1.42.

A bias change translates the boundary along its normal



Signed distance removes arbitrary score scaling

$$d(x) = \frac{w^T x + b}{\|w\|}.$$

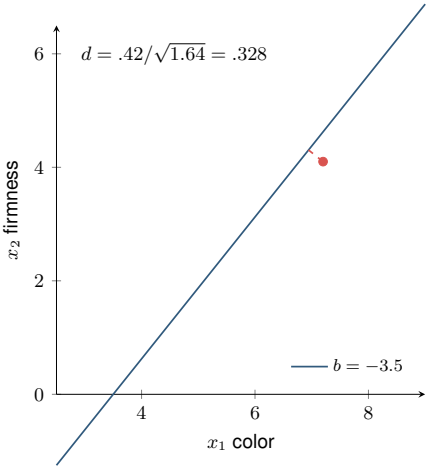
Sorting point. $\|w\| = \sqrt{1 + .8^2} = \sqrt{1.64}$, so

$$d((7.2, 4.1)) = .42/\sqrt{1.64} \approx .328.$$

Parameter change	Raw score	Signed distance
$(w, b) \to (10w, 10b)$	Multiplies by 10	Unchanged
$(w, b) \to (-w, -b)$	Changes sign	Changes sign; class sides swap
$b \to b + \Delta b$	Adds Δb	Boundary moves $-\Delta b/\|w\|$ along w

The distance is geometric in the chosen feature units. It is not automatically a calibrated class probability or a measure invariant to arbitrary feature rescaling.

Signed distance removes arbitrary score scaling



The closest boundary point follows the normal

For a nonzero w, remove the normal component of a point:

$$x_* = x - \frac{w^T x + b}{w^T w} w, \qquad w^T x_* + b = 0.$$

With $x = (7.2, 4.1)$, $g(x) = .42$, and $w = (1, -.8)$,

$$x_* = (7.2, 4.1) - \frac{.42}{1.64}(1, -.8) \approx (6.944, 4.305).$$

$$\|x - x_*\| = \frac{.42}{\sqrt{1.64}} \approx .328.$$

Why it is closest

The displacement is perpendicular to the boundary. Any other boundary point adds a tangential component and therefore a larger squared distance.

This construction checks both the distance formula and the claimed direction of w in one calculation.

Changing units requires changing the coefficients

Original firmness $x_2 = 4.1$ becomes $x_2' = 41$ after a tenfold unit change.

$$x_1 - .8x_2 - 3.5 = x_1 - .08x_2' - 3.5.$$

For standardization $z_j = (x_j - \mu_j)/s_j$,

$$g(x) = \sum_j (w_j s_j) z_j + \left(b + \sum_j w_j \mu_j \right).$$

Example. With $\mu = (6, 5)$ and $s = (2, 1)$, the sorting model becomes

$$g(z) = 2z_1 - .8z_2 - 1.5.$$

Each standardized unit is one training standard deviation. Fit transformation statistics using training data and apply the same transformation to new samples. Coefficient magnitude depends on units; a large raw weight does not by itself establish greater feature importance.

Centering preserves a rule but changes its intercept

Let $z = x - \mu$ with $\mu = (6, 5)^T$. The sorting model becomes

$$g(z) = z_1 - .8z_2 - 1.5, \qquad -1.5 = -3.5 + (1, -.8)^T(6, 5).$$

At the same observation, $z = (1.2, -.9)$ gives

$$1.2 - .8(-.9) - 1.5 = .42.$$

Coefficient interpretation

A one-unit increase in z_1 changes the score by 1 when the other coordinate stays fixed. The intercept is now the score at the chosen feature center.

Centering alone preserves pairwise Euclidean distances. Dividing coordinates by different scales changes the distance geometry, even when coefficients are converted to preserve this particular affine prediction.

Augmentation unifies bias; separability defines capacity

$$\tilde{x} = (x_1, \ldots, x_d, 1)^T, \quad \tilde{w} = (w_1, \ldots, w_d, b)^T, \quad g(x) = \tilde{w}^T \tilde{x}.$$

Linear separability. For labels $y_i \in \{-1, +1\}$, the training set is separable if some parameters satisfy

$$y_i(w^T x_i + b) > 0 \quad \text{for every } i.$$

The last input is always 1, so its weight supplies the intercept. If the bias is omitted, the boundary is forced through the origin and some otherwise separable data become impossible to classify. **Caution.** A separable training sample is a geometric property of these observations. It does not guarantee perfect future prediction, and nonseparable data may require a different representation or loss.

Convex averages give a second separability test

An affine score preserves weighted averages when weights sum to one:

$$g\left(\sum_i \alpha_i x_i\right) = \sum_i \alpha_i g(x_i), \quad \alpha_i \geq 0, \quad \sum_i \alpha_i = 1.$$

For the diagonal-label pattern, the positive corners average to $(0,0)$; the negative corners have the same average.

$$\tfrac{1}{2}(-1,-1) + \tfrac{1}{2}(1,1) = (0,0) = \tfrac{1}{2}(-1,1) + \tfrac{1}{2}(1,-1).$$

Positive corner scores would make $g(0,0) > 0$, while negative corner scores require $g(0,0) < 0$.

General reading

If the convex hulls of two classes share a point, no strict affine separation exists. This geometric obstruction concerns the representation, before any optimizer is chosen.

Worked exercise: recover a rule from a boundary

A boundary passes through $(1, 0)$ and $(0, 2)$. The point $(2, 2)$ must receive the positive label.

1. Find a normal and an intercept without first computing a slope.
2. Evaluate $(0, .5)$ and its signed distance.
3. Give another parameter set that defines the same oriented rule.

Solution

The tangent is $(-1, 2)$, so choose $w = (2, 1)$ and $b = -2$. Then $g(2, 2) = 4 > 0$ confirms the orientation.

$$g(0, .5) = -1.5, \qquad d = -1.5/\sqrt{5} \approx -.671.$$

The parameters $(6, 3, -6)$ multiply the full score by 3, preserving its boundary, class sides, and normalized distances.

XOR shows a limitation of one affine boundary

x_1	x_2	Target y	Required score sign
-1	-1	+1	Positive
-1	+1	-1	Negative
+1	-1	-1	Negative
+1	+1	+1	Positive

Contradiction for $g = w_1x_1 + w_2x_2 + b$. Adding the two positive-corner inequalities gives $2b > 0$. Adding the two negative-corner inequalities gives $2b < 0$. Both cannot hold. This example uses positive labels on equal-sign corners (the complement of the usual XOR label choice); either diagonal pattern is nonseparable. Later, the interaction x_1x_2 will make the pattern simple.

Three ways to combine linear scores into a multiclass rule

Construction	Classifier count	How prediction is made
One versus rest	C	Each class asks whether the input belongs to it
One versus one	$C(C-1)/2$	Pairwise results vote or use another stated combination
Comparable class scores	C	$\hat{y} = \arg\max_k g_k(x)$

Different issues. Independent yes/no rules can overlap or leave gaps. Pairwise votes can form a cycle. Argmax always has a largest score, but tied maxima still need a rule. For 10 classes, OvR uses 10 binary fits and OvO uses 45. Pairwise problems may be easier because each fit sees only two classes; fewer fits do not automatically imply better boundaries.

Fit counts and data exposure answer different cost questions

Suppose $C = 4$ balanced classes contain 30 observations each. Count fits and the total observations presented across fits.

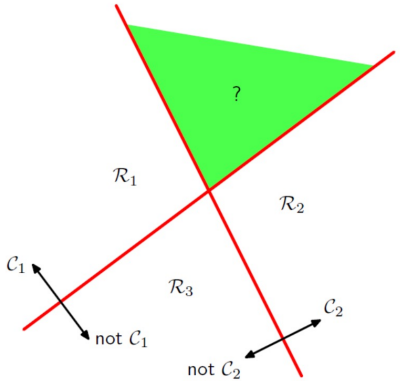
Construction	Fits	Data per fit	Total exposure
OvR	4	120	480
OvO	6	60	360

Each observation participates in C OvR fits and $C - 1$ OvO fits. Actual running time also depends on the fitting algorithm, dimension, parallelism, and class balance.

Geometry still matters

An OvR negative class merges three groups. An OvO fit sees only one opposing group, which can make the binary geometry easier despite requiring more models.

One-versus-rest can leave ambiguous regions



Read the arrows. Each red line separates one class from its rest. The green region is marked “?” because independent binary answers do not supply one agreed class there. Two positive

claims create overlap; no positive claims create a gap. A complete predictor needs a combination policy. Comparable-score argmax resolves the

final choice only when the scores are trained or calibrated for that comparison. Do not compare arbitrary scales from unrelated models as if they were probabilities.

The reference OvR functions exhibit both overlap and a gap

Use the chapter's three binary scores:

$$f_1 = -x_1 + x_2, \quad f_2 = x_1 + x_2 - 5, \quad f_3 = -x_2 + 1.$$

Input	f_1	f_2	f_3	Strict OvR
$(6, 5)$	-1	6	-4	Class 2
$(3, 5)$	2	3	-4	Overlap
$(2, 2)$	0	-1	-1	Gap

Strict OvR accepts a class only when exactly its score is positive. Argmax would choose class 2 at $(3, 5)$ and class 1 at $(2, 2)$.

A changed policy

These argmax outputs complete a decision rule, but comparing independently fitted scales requires justification; completion alone does not establish reliable classification.

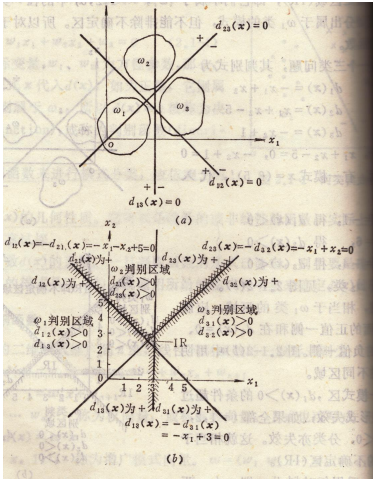
Pairwise votes can agree or cycle

For A, B, C, use three pairwise classifiers.

Pairwise outputs	Votes (A, B, C)	Prediction
A beats B; C beats A; C beats B	$(1, 0, 2)$	C
A beats B; B beats C; C beats A	$(1, 1, 1)$	Tie

Why a cycle is possible. Three separately trained pairwise functions need not be differences of one shared set of class scores. Their preferences can therefore be inconsistent. State a tie policy, such as a fixed class order or a justified margin/probability combination. In a shared-score system, $g_A > g_B$ and $g_B > g_C$ imply $g_A > g_C$, so this strict cycle cannot occur.

Pairwise votes can agree or cycle



The reference pairwise example contains a strict cycle

Let $f_{12} = -x_1 - x_2 + 5$, $f_{13} = -x_1 + 3$, and $f_{23} = -x_1 + x_2$. A positive f_{ab} selects a over b.

Input	f_{12}	f_{13}	f_{23}	Result
$(4, 3)$	-2	-1	-1	Class 3 beats both
$(2.8, 2.5)$	$-.3$	$.2$	$-.3$	One win per class

At the second point, class 2 beats 1, class 1 beats 3, and class 3 beats 2. No class beats every opponent.

Prediction requires a policy

A majority vote also ties here. A declared tie rule can return a label, while the underlying pairwise preferences remain inconsistent.

Shared class scores impose a pairwise consistency identity

If pairwise comparisons come from $f_{ab} = g_a - g_b$, then

$$f_{12} + f_{23} + f_{31} = (g_1 - g_2) + (g_2 - g_3) + (g_3 - g_1) = 0.$$

Three strict preferences $1 > 2$, $2 > 3$, and $3 > 1$ would require all three terms to be positive, contradicting this identity.

Numerical check

For scores $(2, 3, -4)$, the differences are $f_{12} = -1$, $f_{23} = 7$, and $f_{31} = -6$; their sum is zero.

The reference cycle instead has $f_{12} = -.3$, $f_{23} = -.3$, and $f_{31} = -.2$, summing to $-.8$. It therefore cannot be generated by one shared set of class scores at that input.

Argmax compares all classes on one score scale

$$g_1 = -x_1 + x_2, \quad g_2 = x_1 + x_2 - 5, \quad g_3 = -x_2 + 1.$$

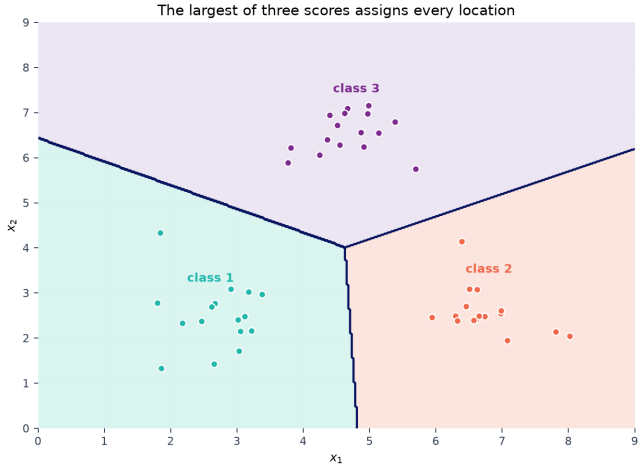
Input	g_1	g_2	g_3	Argmax
$(6, 5)$	-1	6	-4	Class 2
$(3, 5)$	2	3	-4	Class 2

At $(3, 5)$, two scores are positive, so strict OvR would be ambiguous; argmax still selects the larger score.

$$g_c(x) = g_k(x) \quad \Longrightarrow \quad (w_c - w_k)^T x + (b_c - b_k) = 0.$$

Class c receives the intersection of all inequalities $g_c \geq g_k$. The visible class region is therefore an intersection of half-spaces, with ties on its boundaries.

Argmax compares all classes on one score scale



An affine argmax class occupies a convex region

For the shared scores $g_1 = -x_1 + x_2$, $g_2 = x_1 + x_2 - 5$, and $g_3 = -x_2 + 1$, class 1 requires

$$g_1 \geq g_2 \Rightarrow x_1 \leq 2.5, \qquad g_1 \geq g_3 \Rightarrow 2x_2 - x_1 \geq 1.$$

Its region is the intersection of these half-spaces. Points $(0, 1)$ and $(2, 2)$ both satisfy the inequalities; their midpoint $(1, 1.5)$ does too.

Capacity consequence

Every segment between two points assigned to this region stays inside it, apart from the stated boundary tie policy. One affine score per class therefore cannot represent two isolated class islands separated by another class.

Multiple prototypes later provide unions of local regions.

Common score transformations preserve argmax

At $(3, 5)$, the shared scores are $(2, 3, -4)$ and class 2 wins.

Transformation	New scores	Largest class
Add 10 to every score	$(12, 13, 6)$	2
Multiply every score by 2	$(4, 6, -8)$	2
Multiply only g_1 by 2	$(4, 3, -4)$	1

More generally, $g'_c(x) = a(x)g_c(x) + h(x)$ preserves the winning set when $a(x) > 0$ and both terms are common across classes.

Use in derivations

Class-independent terms may be removed during score comparison. Class-specific offsets or scales modify the differences and can move decision boundaries.

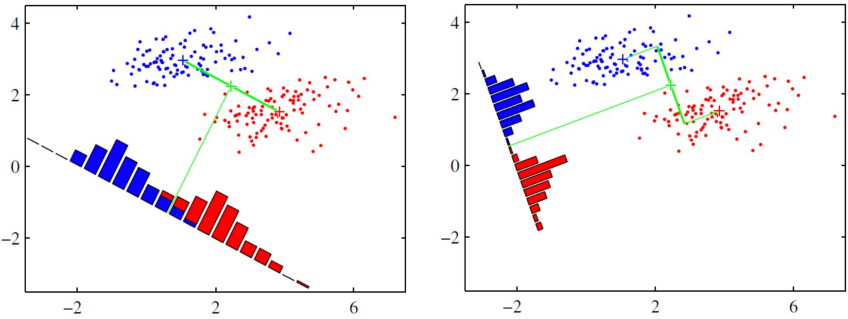
Exercise: preserve the rule, then interpret the score

Use the sorting score $g = x_1 - .8x_2 - 3.5$.

1. At $x = (7.2, 4.1)$, calculate the score, label, and signed distance.
2. Firmness is now recorded as $x_2' = 10x_2$. Give equivalent coefficients.
3. Multiply all equivalent coefficients by 5. Which quantities change?
4. Is a score .42 a 42% chance of a ripe tomato? Explain.

Check after discussion. Score .42, positive label, distance .328 in the original feature geometry. Equivalent new firmness coefficient: $-.08$. Positive parameter scaling changes raw scores but preserves the boundary and labels. A probability requires a specified probability model, not merely a score.

Projection can keep or destroy class separation



Compare the projections. Both panels show the same blue/red observations. The projection axis changes. Histograms along the axis show how the two classes overlap after projection. Fisher asks for class centers to be separated relative to within-class spread. A long center-to-center distance alone is insufficient when each projected class is also wide. The original points have not moved.

A dot product gives a one-dimensional coordinate

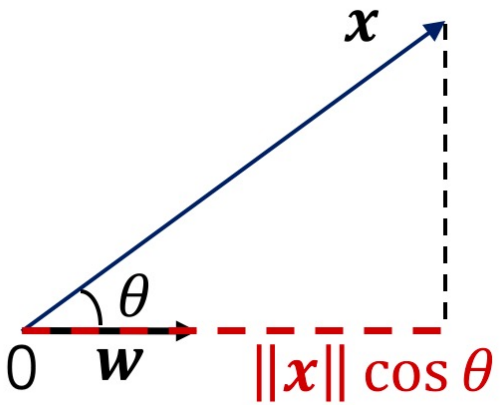
$$z = w^T x.$$

For unit w, $z = \|x\| \cos\theta$ is the signed coordinate along the projection axis. The projected vector in the original space is zw.

Point	$w = (1,0)^T$	$w = (0,1)^T$	$w = (1,1)^T/\sqrt{2}$
$(2,0)$	2	0	$\sqrt{2}$
$(2,2)$	2	2	$2\sqrt{2}$

The horizontal projection loses the difference between these two points; the vertical or diagonal projection retains it. Fisher chooses a direction using labels, rather than selecting the direction with the greatest overall variance. The latter is the different question asked by PCA.

A dot product gives a one-dimensional coordinate



Class statistics separate location from within-class variation

$$m_k = \frac{1}{n_k} \sum_{i:y_i=k} x_i, \qquad S_k = \sum_{i:y_i=k} (x_i - m_k)(x_i - m_k)^T.$$

$$S_W = S_1 + S_2, \qquad S_B = (m_1 - m_2)(m_1 - m_2)^T.$$

Quantity	What it measures
$m_1 - m_2$	Separation of the two class centers
S_W	Summed deviations inside each class
S_B	Center separation written as a quadratic form

Scatter is a sum, not an average covariance. Center samples by their own class means. Centering everything by the grand mean would mix between-class separation into the denominator we intend to keep small.

Scatter and covariance differ by a stated denominator

Our eight-point dataset has $S_W = \begin{bmatrix} 4 & -2 \\ -2 & 2 \end{bmatrix}$.

Convention	Denominator	Pooled matrix
Within-class scatter	None	S_W
Gaussian MLE covariance	$n = 8$	$S_W/8$
Unbiased pooled covariance	$n - C = 6$	$S_W/6$

$$S_W/8 = \begin{bmatrix} .5 & -.25 \\ -.25 & .25 \end{bmatrix}, \qquad S_W/6 = \begin{bmatrix} 2/3 & -1/3 \\ -1/3 & 1/3 \end{bmatrix}.$$

Replacing S_W by a common positive multiple preserves the Fisher direction but rescales its coefficients and objective value. A probabilistic threshold must use the covariance and score scale from the same fitted model; naming every matrix merely “variance” hides this distinction.

Total scatter includes the separation between class centers

For the same eight points, the grand mean is $m = (2, 1.5)^T$. Define the weighted between-class scatter

$$S_B^{(w)} = \sum_{k=1}^{2} n_k (m_k - m)(m_k - m)^T = \begin{bmatrix} 0 & 0 \\ 0 & 8 \end{bmatrix}.$$

Then

$$S_T = \sum_i (x_i - m)(x_i - m)^T = S_W + S_B^{(w)} = \begin{bmatrix} 4 & -2 \\ -2 & 10 \end{bmatrix}.$$

Derivation

Write $x_i - m = (x_i - m_k) + (m_k - m)$. The cross terms vanish because each class-centered sum is zero.

The earlier binary convention was $S_B = \Delta\Delta^T = \mathrm{diag}(0, 4)$. Here $S_B^{(w)}$ uses sample counts. Keep their definitions distinct even though proportional matrices yield the same binary Fisher direction.

Projected means and scatter follow directly from the dot product

$$\tilde{m}_k = \frac{1}{n_k} \sum_{i:y_i=k} w^T x_i = w^T m_k,$$

$$\tilde{S}_k = \sum_{i:y_i=k} (w^T x_i - w^T m_k)^2 = w^T S_k w,$$

$$(\tilde{m}_1 - \tilde{m}_2)^2 = [w^T (m_1 - m_2)]^2 = w^T S_B w.$$

Dimensions. $m_k, w \in \mathbb{R}^d$, $S_k \in \mathbb{R}^{d \times d}$, but all projected quantities above are scalars. No new cloud needs to be generated to compare directions: the original class statistics summarize the relevant projected means and summed squared deviations. These equalities convert the geometric goal into one objective involving w.

Fisher measures separation relative to within-class spread

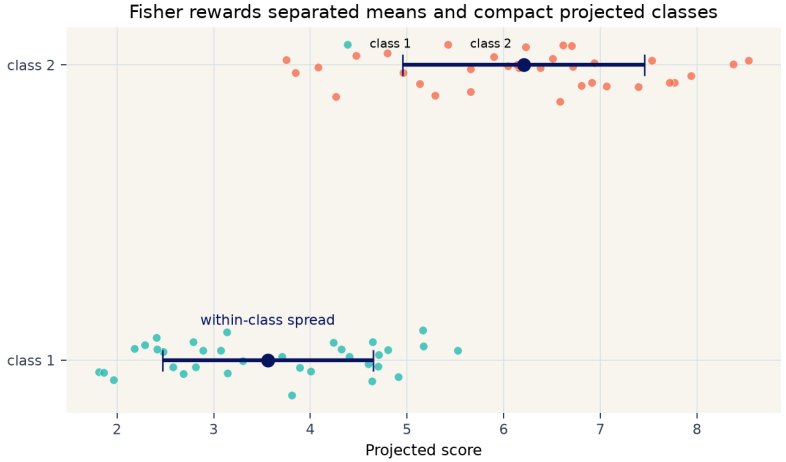
$$J(w) = \frac{[w^T(m_1 - m_2)]^2}{w^T S_W w} = \frac{w^T S_B w}{w^T S_W w}.$$

Part	Desired behavior
Numerator	Projected class centers far apart
Denominator	Projected observations compact within each class
Ratio	Strong class separation relative to class variation

$$J(cw) = J(w) \quad \text{for nonzero } c.$$

Scaling the axis stretches both the center gap and spread, leaving their ratio unchanged. For a derivation, fix $w^T S_W w = 1$. For display, normalize $\|w\| = 1$. The criterion chooses a direction. It does not by itself specify class priors, error costs, or a final threshold.

Fisher measures separation relative to within-class spread



A noisy mean-difference direction can be inferior

Suppose $\Delta = m_1 - m_2 = (2,1)^T$ and $S_W = \mathrm{diag}(16,1)$.

Direction w	$(w^T\Delta)^2$	$w^T S_W w$	$J(w)$
Mean difference $(2,1)$	25	65	.385
Vertical $(0,1)$	1	1	1
Fisher $(1/8,1)$	$25/16$	$5/4$	1.25

The first coordinate separates the means but varies greatly within classes. Fisher downweights that noisy direction without throwing it away completely. More geometric center separation does not guarantee a

better projection. Compare the center gap with the spread measured in the same direction, using the same scatter convention.

Candidate directions reveal where the Fisher gain comes from

For $\Delta = (0, -2)^T$ and $S_W = \begin{bmatrix} 4 & -2 \\ -2 & 2 \end{bmatrix}$,

$$J(w) = \frac{4w_2^2}{4w_1^2 - 4w_1 w_2 + 2w_2^2}.$$

Direction	Squared mean gap	Within scatter	Ratio
$(1, 0)$	0	4	0
$(0, 1)$	4	2	2
$(1, 1)$	4	2	2
$(1, 2)$	16	4	4
$(1, 3)$	36	10	3.6

The negative cross term reduces projected within-class scatter when both weights have the same sign. Increasing the second weight beyond the optimum increases the mean gap, but the denominator grows enough to reduce the ratio.

Derive the direction from a constrained objective

Maximize $w^T S_B w$ subject to $w^T S_W w = 1$:

$$\mathcal{L}(w, \lambda) = w^T S_B w - \lambda(w^T S_W w - 1).$$

For symmetric S_B, S_W,

$$\nabla_w \mathcal{L} = 2S_B w - 2\lambda S_W w = 0 \quad \Longrightarrow \quad S_B w = \lambda S_W w.$$

Let $\Delta = m_1 - m_2$. Because $S_B = \Delta\Delta^T$,

$$S_B w = \Delta(\Delta^T w), \qquad w \propto S_W^{-1}\Delta.$$

Assumptions. Distinct class means and positive-definite S_W give the nonzero separating solution. The constant sets scale, not direction. Numerically, solve $S_W w = \Delta$; do not explicitly form the inverse.

Cauchy–Schwarz also derives the optimal direction

Assume S_W is positive definite. Let $v = S_W^{1/2} w$ and $a = S_W^{-1/2} \Delta$. Then

$$J(w) = \frac{(v^T a)^2}{v^T v} \leq a^T a = \Delta^T S_W^{-1} \Delta.$$

Equality in Cauchy–Schwarz holds when v is proportional to a, giving

$$w \propto S_W^{-1} \Delta.$$

For our dataset, $S_W^{-1} \Delta = (-1, -2)^T$ and the upper bound is

$$(0, -2)^T (-1, -2) = 4.$$

What this proves

The earlier direction is globally optimal for this positive-definite binary criterion, rather than merely a stationary point. Singular scatter requires separate treatment; the inverse square root used here would not exist.

Interpret the inverse scatter as a directional correction

$$w \propto S_W^{-1}(m_1 - m_2).$$

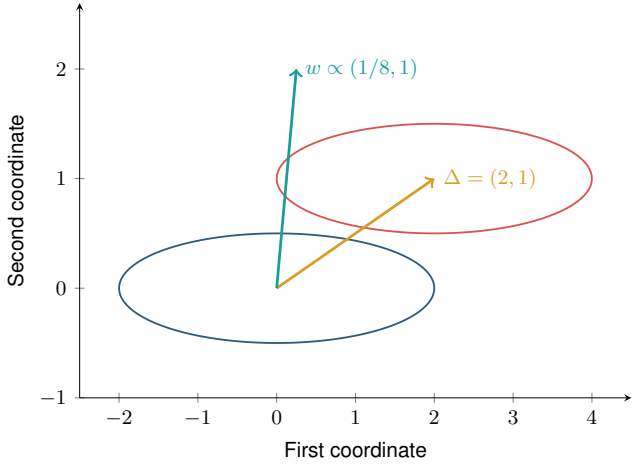
If $S_W = \mathrm{diag}(s_1, s_2)$,

$$w_j \propto \frac{m_{1j} - m_{2j}}{s_j}.$$

Meaning. The same center difference contributes less on a direction with high within-class variation. Correlated features require a joint matrix correction, not independent division by marginal variances. **Example.**

For $\Delta = (2, 1)$ and $S_W = \mathrm{diag}(16, 1)$, solving gives $(1/8, 1)$. Multiplying by a negative constant reverses projected ordering, so the threshold inequality must also reverse. Normalizing to unit length is convenient for plotting but does not change the Fisher ratio or class information.

Interpret the inverse scatter as a directional correction



Whitening makes the directional correction visible

Consider $\Delta = (2,1)^T$ and $S_W = \mathrm{diag}(16,1)$. Transform coordinates by

$$u = S_W^{-1/2} x = (x_1/4, x_2)^T.$$

The transformed within-class scatter is the identity, and the transformed mean difference is $\Delta_u = (.5,1)^T$. In this geometry, choose a direction parallel to Δ_u.

$$\Delta_u^T u = .5(x_1/4) + x_2 = (1/8)x_1 + x_2.$$

Original-space direction

Mapping the score back gives $w = (1/8,1)^T$, exactly the Fisher direction.

The first coordinate is useful, but its large within-class spread makes it less reliable for discrimination. Whitening rescales the entire within-class geometry; it is more general than dividing only by marginal standard deviations when correlations are present.

Training data for one complete Fisher calculation

Class	Points	Mean
1	$(1,1),(2,0),(2,1),(3,0)$	$(2,.5)^T$
2	$(1,3),(2,2),(2,3),(3,2)$	$(2,2.5)^T$

$$\Delta = m_1 - m_2 = (0,-2)^T.$$

What the data suggest. The second coordinate changes between classes; both classes share the same within-class shape because class 2 is a translation of class 1 by $(0,2)$. **Do next.** Center each class separately, accumulate outer products, solve $S_W w = \Delta$, project the observations, and choose a threshold. A mean difference alone is insufficient because the features also vary together.

Compute the class scatter from four centered rows

Class-1 deviation	δ_1^2	$\delta_1\delta_2$	δ_2^2
$(-1, .5)$	1	-.5	.25
$(0, -.5)$	0	0	.25
$(0, .5)$	0	0	.25
$(1, -.5)$	1	-.5	.25
Sum	2	-1	1

$$S_1 = S_2 = \begin{bmatrix} 2 & -1 \\ -1 & 1 \end{bmatrix}, \qquad S_W = \begin{bmatrix} 4 & -2 \\ -2 & 2 \end{bmatrix}.$$

The off-diagonal entry is negative because horizontal and vertical deviations tend to have opposite signs.

Scatter sums squared deviations; it is not divided by 4 in the Fisher convention used here.

Solve the Fisher system and compare the resulting ratio

$$\begin{bmatrix} 4 & -2 \\ -2 & 2 \end{bmatrix} \begin{bmatrix} w_1 \\ w_2 \end{bmatrix} = \begin{bmatrix} 0 \\ -2 \end{bmatrix}.$$

The first equation gives $w_2 = 2w_1$; the second gives $w_1 = -1$, hence

$$w = (-1, -2)^T.$$

Direction	Squared mean gap	Projected within scatter	J
Vertical $(0, -1)$	4	2	2
Fisher $(-1, -2)$	16	4	4

The Fisher direction uses the negative within-class correlation to reduce projected variation relative to the gap. It improves the ratio even though the class means differ only vertically.

The chapter's isotropic example contrasts with correlated scatter

The reference example gives $m_1 = (2,0)^T$, $m_2 = (2,2)^T$, and $S_W = 2I$.

$$w = (2I)^{-1}(0,-2)^T = (0,-1)^T, \qquad t = \tfrac{1}{2}w^T(m_1 + m_2) = -1.$$

The boundary is $-x_2 = -1$, or $x_2 = 1$, with class 1 below it.

Example	Within-class geometry	Fisher normal
Reference page 30	$2I$, no preferred noise direction	$(0,-1)$
Our eight points	Correlated coordinates	$(-1,-2)$

Both mean differences are vertical. Only the second scatter contains correlation, so only that example gains from a tilted projection. The center difference alone cannot determine the final direction.

Projection needs a threshold and a class-side rule

With $w = (-1, -2)^T$:

Class	Projected training values $z = w^T x$	Projected mean
1	$-3, -2, -4, -3$	-3
2	$-7, -6, -8, -7$	-7

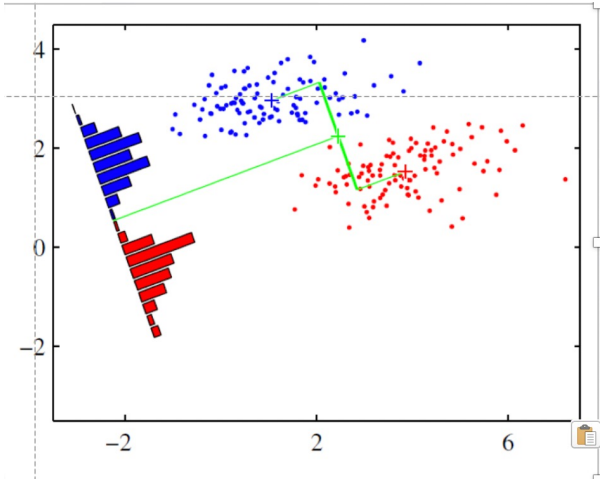
$$t = \frac{-3 + (-7)}{2} = -5.$$

Equal-prior midpoint rule. Predict class 1 if $z > -5$, class 2 if $z < -5$. The boundary is

$$-x_1 - 2x_2 = -5 \quad \Longleftrightarrow \quad x_1 + 2x_2 = 5.$$

For $x = (2, 1)$, $z = -4$ and predict class 1. For $x = (2, 2)$, $z = -6$ and predict class 2. Choosing a direction and choosing a threshold are distinct steps.

Projection needs a threshold and a class-side rule



Axis scaling and sign changes require matching thresholds

Our original rule is $w = (-1, -2)$, $t = -5$, and class 1 when $w^T x > t$.

Axis choice	New threshold	Class-1 inequality
$w/\sqrt{5}$	$-5/\sqrt{5} = -\sqrt{5}$	$w^T x/\sqrt{5} > -\sqrt{5}$
$10w$	-50	$10w^T x > -50$
$-w$	5	$-w^T x < 5$

For $x = (2, 1)$, the original projection is -4. Under axis reversal it becomes 4, which is still class 1 because $4 < 5$.

Separate invariances

The Fisher ratio is unchanged by any nonzero scaling. A prediction rule is preserved only when the threshold and, for negative scaling, the inequality direction are changed consistently.

Priors and losses set the projected threshold

Assume projected models $z \mid k \sim \mathcal{N}(\mu_k, \sigma_z^2)$ with equal variance. Then

$$\log \frac{P(1 \mid z)}{P(2 \mid z)} = \frac{\mu_1 - \mu_2}{\sigma_z^2} \left(z - \frac{\mu_1 + \mu_2}{2} \right) + \log \frac{\pi_1}{\pi_2}.$$

For miss-class-1 cost c_{21} and false-class-1 cost c_{12}, select class 1 when posterior odds exceed c_{12}/c_{21}.

$$t = \frac{\mu_1 + \mu_2}{2} + \frac{\sigma_z^2}{\mu_1 - \mu_2} \log \frac{c_{12}\pi_2}{c_{21}\pi_1}.$$

Our fitted example. $\mu_1 = -3$, $\mu_2 = -7$, $\sigma_z^2 = .5$. Equal costs and $\pi_1 = .8$ give $t = -5 - \frac{1}{8} \log 4 \approx -5.173$. Class 1 gains a larger region.

A costly missed class moves the cut on a fixed projection

Keep the illustrative projected Gaussian fit $\mu_1 = -3$, $\mu_2 = -7$, $\sigma_z^2 = .5$, and equal priors. Let a missed class-1 observation cost $c_{21} = 5$, while a false class-1 decision costs $c_{12} = 1$.

$$t = -5 + \frac{.5}{4} \log \frac{1}{5} = -5 - \frac{\log 5}{8} \approx -5.201.$$

At $z = -5.1$, the equal-cost midpoint rule predicts class 2; the costly-miss rule predicts class 1.

Interpretation

A lower threshold expands the class-1 region because class 1 has the larger projected mean.

The direction and class-side rule stay fixed. This numerical shift uses the additional equal-variance Gaussian assumption; the Fisher scatter ratio alone does not provide a loss-sensitive threshold.

Fisher and Gaussian LDA share a direction under conditions

Method	What is assumed/optimized	What is obtained
Fisher projection	Maximize mean gap relative to within scatter	Direction $S_W^{-1}(m_1 - m_2)$
Gaussian LDA	Gaussian classes with one shared covariance	Log-odds direction $\Sigma^{-1}(\mu_1 - \mu_2)$ plus intercept

If the pooled within-class covariance is proportional to S_W, the estimated directions agree up to scale. Gaussian LDA additionally supplies class probabilities under its distributional model. Fisher's criterion alone is not a Gaussian likelihood and does not guarantee calibrated posterior probabilities. Midpoint thresholds, class priors, and losses must be stated separately. Multimodal projected classes can still overlap despite a large center gap.

Regularization limits amplification of weak directions

With two classes and n total samples,

$$\operatorname{rank}(S_W) \leq \min(d, n-2).$$

Duplicated or constant features can reduce the rank further. Replace the singular or unstable system by

$$(S_W + \lambda I) w_\lambda = m_1 - m_2, \qquad \lambda > 0.$$

Situation	Effect of regularization
Nearly zero scatter direction	Prevents an arbitrarily large inverse weight
Larger λ	Pulls the direction toward a scaled mean difference
Features with incompatible units	Isotropic penalty changes meaning; preprocessing matters

Choose the regularization strength on validation data. It adds a modelling preference for stable directions; it does not create new evidence about unobserved variation.

Regularization changes the direction in a measurable way

Use the eight-point scatter and add λI with $\lambda = 2$:

$$\begin{bmatrix} 6 & -2 \\ -2 & 4 \end{bmatrix} w_\lambda = \begin{bmatrix} 0 \\ -2 \end{bmatrix}, \qquad w_\lambda = (-.2, -.6)^T.$$

Projected class means are $-.7$ and -1.9, so the midpoint threshold is -1.3.

$$J(w_\lambda) = \frac{1.2^2}{.4} = 3.6 \quad \text{using the original, unregularized } S_W.$$

Tradeoff

The original direction has $J = 4$. Regularization deliberately sacrifices some training criterion value in favor of reduced sensitivity to weak scatter directions.

This dataset is already invertible; the calculation illustrates the effect rather than establishing that $\lambda = 2$ improves prediction. Compare validation performance before choosing its value.

Equal class means can defeat the Fisher mean-gap criterion

Consider four class-1 points $(\pm 1, 0), (0, \pm 1)$ and four class-2 points $(\pm 3, 0), (0, \pm 3)$.

$$m_1 = m_2 = (0, 0), \qquad S_W = 20I, \qquad \Delta = 0.$$

Every nonzero direction has $J(w) = 0$: both projected means coincide even though the radius perfectly separates these eight observations.

$$r^2 = x_1^2 + x_2^2, \qquad r^2 = 1 \text{ for class 1}, \quad r^2 = 9 \text{ for class 2}.$$

Representation diagnosis

A score $g = 5 - r^2$ is positive on class 1 and negative on class 2. The useful difference is radial structure, which a projected mean-gap objective does not summarize.

This is a stated classroom construction, not evidence that every real concentric dataset is noise-free.

The program mirrors means, scatter, direction, and threshold

```
import numpy as np
X1 = np.array([[1.,1.],[2.,0.],[2.,1.],[3.,0.]])
X2 = X1 + np.array([0.,2.])
m1, m2 = X1.mean(0), X2.mean(0)
C1, C2 = X1-m1, X2-m2
Sw = C1.T @ C1 + C2.T @ C2
w = np.linalg.solve(Sw, m1-m2)
z1, z2 = X1 @ w, X2 @ w
t = .5*(z1.mean()+z2.mean())
print(Sw, w, t)  # [[4,-2],[-2,2]], [-1,-2], -5
```

Inspect the outputs. C_k holds class-centered rows; $C_k^T C_k$ is the scatter. Prediction compares $x^T w$ with t using the class-side rule. Add λI to the solve when regularization is required, keeping the chosen convention explicit.

Multiclass Fisher uses weighted class-center scatter

For C classes, define $n = \sum_k n_k$, $m = \sum_k n_k m_k / n$, and

$$S_W = \sum_k S_k, \qquad S_B = \sum_k n_k (m_k - m)(m_k - m)^T, \qquad S_T = S_W + S_B.$$

Three-class calculation

Take $n_1 = n_2 = n_3 = 2$ and centers $(0,0)$, $(3,0)$, $(0,3)$. Their grand mean is $(1,1)$.

$$S_B = 2 \left\{ \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} + \begin{bmatrix} 4 & -2 \\ -2 & 1 \end{bmatrix} + \begin{bmatrix} 1 & -2 \\ -2 & 4 \end{bmatrix} \right\} = \begin{bmatrix} 12 & -6 \\ -6 & 12 \end{bmatrix}.$$

Each center is weighted by its class size. The binary center-difference outer product is replaced by scatter around the shared grand mean, so several independent separation directions may now carry class information.

The number of Fisher axes is bounded by class count

Let A contain the columns $\sqrt{n_k}(m_k - m)$. Then $S_B = AA^T$, and

$$\sum_k n_k(m_k - m) = 0.$$

The C center deviations have a linear dependence, giving

$$\operatorname{rank}(S_B) \le \min(d, C-1), \qquad K \le \min(d, C-1).$$

Classes and features	Maximum useful axes	Interpretation
$C = 2, d = 20$	1	One mean-separation direction
$C = 3, d = 20$	2	Two independent center directions
$C = 8, d = 3$	3	Original dimension also limits rank

Collinear or repeated class centers can reduce the rank further. The bound counts nonzero center-separation directions; it does not guarantee that projecting onto all of them eliminates class overlap.

A normalized multiclass objective gives generalized eigenvectors

Assume S_W is positive definite. If regularization is needed, use the declared regularized matrix consistently. Choose a $d \times K$ projection matrix W and impose

$$W^T S_W W = I_K.$$

Maximize $\mathrm{tr}(W^T S_B W)$ under this constraint. Its columns can be selected from the largest generalized eigenvalues:

$$S_B w_j = \lambda_j S_W w_j.$$

Our three-class centers

If the stated within-class scatter is $S_W = I_2$, the matrix $S_B = \begin{bmatrix} 12 & -6 \\ -6 & 12 \end{bmatrix}$ has eigenvalues 18 and 6.

$$w_1 = (1, -1)^T/\sqrt{2}, \qquad w_2 = (1, 1)^T/\sqrt{2}.$$

One axis captures the largest normalized center separation; retaining both captures the rank-two center geometry. This constrained trace objective must not be confused with an unrestricted ratio of two traces, which generally has different optimization properties.

Projected centers still require a classification rule

With $w_1 = (1, -1)/\sqrt{2}$ and $w_2 = (1, 1)/\sqrt{2}$, set $z = W^T x$.

Original center	First coordinate	Second coordinate
$(0, 0)$	0	0
$(3, 0)$	$3/\sqrt{2}$	$3/\sqrt{2}$
$(0, 3)$	$-3/\sqrt{2}$	$3/\sqrt{2}$

For $x = (2.5, .2)$, $z = (2.3, 2.7)/\sqrt{2}$. With both orthonormal axes, squared distances to the centers remain 6.29, .29, and 14.09; nearest-center prediction is class 2.

Two operations

Fisher learns a label-informed representation. A classifier then turns projected coordinates into labels.

Gaussian LDA can provide both operations under its probabilistic model; a transform alone does not specify probabilities or the final decision policy.

Explain Fisher in a calculation, not just a formula

Short exercise. Given $m_1 = (0,0)^T$, $m_2 = (2,1)^T$, $S_W = \mathrm{diag}(8,2)$:

1. Solve $S_W w = m_1 - m_2$.
2. Project both means and give the midpoint threshold.
3. Classify $x = (1,0)^T$ using the correct class-side rule.
4. Explain why multiplying w and the threshold by 10 changes no labels.

Solution. $w = (-1/4, -1/2)^T$; projected means $0, -1$; $t = -1/2$. New $z = -1/4 > t$, so class 1. The criterion is scale-invariant, but prediction still needs a threshold. Changing priors or losses can move that threshold while keeping the chosen projection direction.

A fixed feature map makes nonlinear structure accessible

$$\phi(x) = (\phi_1(x), \ldots, \phi_M(x))^T, \qquad g(x) = w^T \phi(x) + b.$$

Original observation	Added feature	New structure represented
One measurement x	x^2	Two threshold crossings
Two signs (x_1, x_2)	$x_1 x_2$	Sign agreement/XOR pattern
Color and firmness	Interaction term	Effect of color depends on firmness

Linear **in parameters** does not imply linear **in the original input**. The feature map is chosen before fitting w. Its interpretation depends on the measurement definition and preprocessing. This "generalized linear discriminant" means a linear score on fixed features; it is not the same terminology as the statistical GLM family.

A quadratic score isolates a middle interval

Desired rule. Class 2 for $1 < x < 3$; class 1 outside that interval.

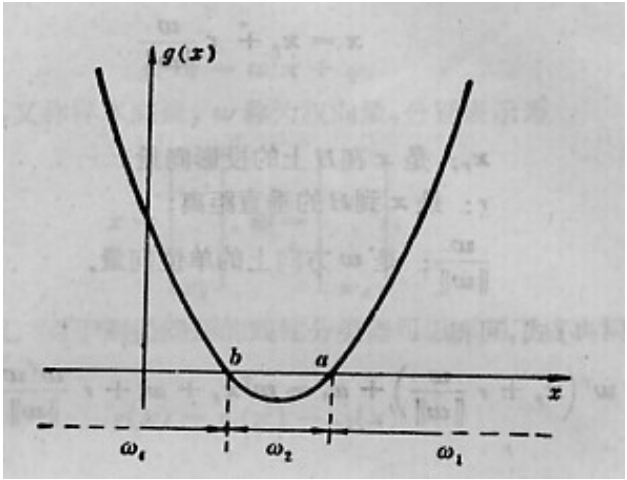
$$g(x) = (x-1)(x-3) = x^2 - 4x + 3.$$

$$\phi(x) = (x^2, x, 1)^T, \qquad w = (1, -4, 3)^T.$$

Input	Score	Class
0	3	1
1.5	-.75	2
2	-1	2
3.5	1.25	1

One affine function of x has at most one zero crossing. This score has two. In lifted coordinates $(z_1, z_2) = (x^2, x)$, the boundary $z_1 - 4z_2 + 3 = 0$ is a line; the original inputs lie on $z_1 = z_2^2$.

A quadratic score isolates a middle interval



Two roots encode an inside–outside interval rule

For endpoints $a < b$, use

$$g(x) = (x - a)(x - b) = x^2 - (a + b)x + ab.$$

The score is negative between the roots and positive outside. For $a = 2, b = 5$,

$$g = x^2 - 7x + 10, \qquad \phi = (x^2, x, 1), \quad w = (1, -7, 10).$$

Input	Product	Region
1	$(-1)(-4) = 4$	Outside
3	$(1)(-2) = -2$	Inside
6	$(4)(1) = 4$	Outside

In lifted coordinates $(z_1, z_2) = (x^2, x)$, inputs follow a parabola. The line $z_1 - 7z_2 + 10 = 0$ crosses it at the two roots. Endpoint equality still needs a declared label policy.

One interaction resolves the diagonal-label pattern

x_1	x_2	$x_1 x_2$	Target
-1	-1	+1	+1
-1	+1	-1	-1
+1	-1	-1	-1
+1	+1	+1	+1

$$g(x) = x_1 x_2, \qquad \hat{y} = \operatorname{sign} g(x).$$

What changed? The new feature records sign agreement, which no weighted sum of the original two coordinates can represent on these four corners. All equal-sign points now have one feature value and all opposite-sign points another. This solves the stated training geometry exactly. It does not claim that an arbitrary real dataset becomes easy after adding every possible interaction.

Translating coordinates changes the interaction expansion

The score $g = x_1 x_2$ expresses the diagonal-label pattern around the original origin. Translate to $u = x_1 - 1$ and $v = x_2 + 2$:

$$x_1 = u + 1, \quad x_2 = v - 2, \qquad g = (u+1)(v-2) = uv - 2u + v - 2.$$

For $(x_1, x_2) = (2, 1)$, $(u, v) = (1, 3)$ and

$$uv - 2u + v - 2 = 3 - 2 + 3 - 2 = 2 = x_1 x_2.$$

Feature choice

Keeping only the translated interaction uv would define a different classifier. Linear and constant terms are needed to preserve this same original score.

Centering before polynomial expansion is a modelling choice. A complete quadratic basis can represent the translated rule; an interaction-only basis may not remain equivalent under a change of origin.

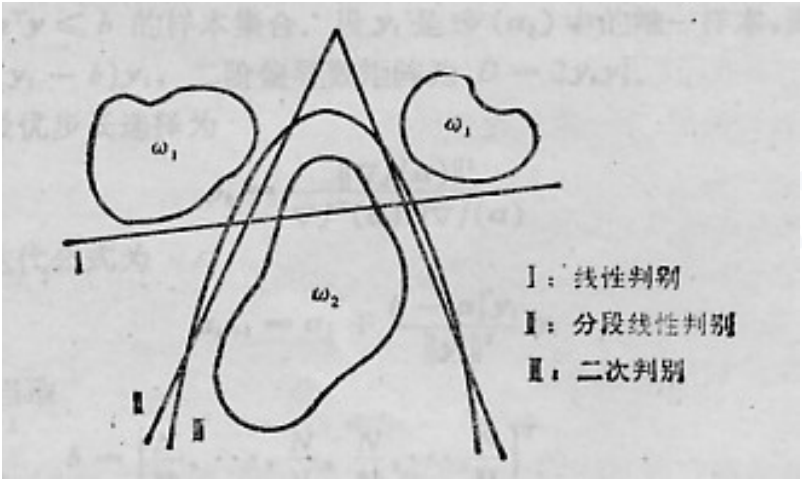
A full quadratic map expresses several boundary shapes

$$\phi(x) = (x_1^2, x_1x_2, x_2^2, x_1, x_2, 1)^T.$$
$$g(x) = ax_1^2 + bx_1x_2 + cx_2^2 + dx_1 + ex_2 + f.$$

Score equation $g(x) = 0$	Example geometry
$x_1^2 + x_2^2 - 4 = 0$	Circle of radius 2
$x_1^2 - x_2 = 0$	Parabola
$x_1x_2 = 0$	Two intersecting lines

For input $(1, 2)$, the feature vector is $(1, 2, 4, 1, 2, 1)$. Different weights select different conic or degenerate shapes. The model remains linear in its six coefficients; adding features changes capacity before any fitting procedure is chosen.

A full quadratic map expresses several boundary shapes



A quadratic matrix counts the cross coefficient twice

A quadratic score can be written as $g = x^T Qx + a^T x + b$ with symmetric Q.

$$x^T Qx = Q_{11}x_1^2 + 2Q_{12}x_1x_2 + Q_{22}x_2^2.$$

For $g = x_1^2 + 2x_1x_2 + 3x_2^2 - 4x_1 + 5$,

$$Q = \begin{bmatrix} 1 & 1 \\ 1 & 3 \end{bmatrix}, \quad a = (-4, 0)^T, \quad b = 5.$$

At $x = (1, 2)$, the feature-vector calculation gives

$$1 + 2(2) + 3(4) - 4 + 5 = 18.$$

The matrix calculation gives $x^T Qx = 17$, then adds $-4 + 5$ to obtain 18.

Common mismatch

Putting the displayed interaction coefficient 2 into both off-diagonal entries produces $4x_1x_2$. The monomial coefficient and symmetric-matrix entry are different conventions.

Polynomial expansion trades flexibility for parameter growth

$$M_{\text{degree}\le r} = \binom{d+r}{r}, \qquad M_{\text{degree}\le 2} = \frac{(d+1)(d+2)}{2}.$$

Input dimension	Degree	Feature count	Practical question
2	2	6	Which interactions are meaningful?
10	3	286	Is there enough labelled support?
100	2	5,151	Can we regularize and compute reliably?
100	3	176,851	Is this expansion justified?

Squaring scales near 10 versus .1 changes them to 100 versus .01. Standardization and regularization therefore matter even more after expansion. A richer map may fit sampling noise between observed points; choose its degree using validation performance, not training fit alone.

Polynomial feature options determine which terms exist

For three inputs, a complete degree-two basis including the constant has ten terms:

$$1,\ x_1, x_2, x_3,\ x_1^2, x_1x_2, x_1x_3, x_2^2, x_2x_3, x_3^2.$$

For $x = (1, 2, 3)$, these become

$$(1, 1, 2, 3, 1, 2, 3, 4, 6, 9).$$

Configuration	Term count	Missing terms
Complete quadratic, bias included	10	None
Interaction only, bias included	7	x_1^2, x_2^2, x_3^2
Complete quadratic, no bias	9	Constant 1

If the classifier already fits an intercept, adding a constant feature duplicates that role. Record feature order as well as options, because each fitted coefficient must be paired with the correct monomial at prediction time.

A basis should express the structure we actually need

Basis	What it expresses	Main choice
Polynomial	Broad smooth global variation	Degree and selected interactions
Radial basis	Similarity near a center	Centers and widths
Spline	Smooth connected local pieces	Knot placement
Domain interaction	A known relation among measurements	Meaning after preprocessing

Reasoning example. A middle interval can use $(x-1)(x-3)$. Two spatially separated class clusters may be easier to represent by several prototypes than by a high-degree polynomial. Separate representation choice from weight fitting. Prefer enough flexibility to express repeated structure, with a controlled way to evaluate its performance on unseen observations.

A periodic feature respects a circular measurement

An hour-of-day feature has a wrap-around: hours 23 and 1 are two hours apart around midnight, although their raw numerical difference is 22.

$$\phi(h) = \left(\cos\frac{2\pi h}{24}, \sin\frac{2\pi h}{24}\right).$$

For a midnight-centered rule, use $g(h) = \cos(2\pi h/24) - .5$.

$$g(23) = g(1) \approx .466, \qquad g(12) = -1.5.$$

Representation meaning

The two coordinates encode location on a cycle; a linear score in these coordinates can select a time window that crosses the original numerical endpoint.

This classroom example illustrates the chapter's general fixed-function mapping. Choose a basis from the measurement structure, rather than assuming higher polynomial degree is always the appropriate response.

Least squares fits targets but can overemphasize extreme scores

Encode labels by $t_i \in \{-1, +1\}$ and include a constant column in Φ:

$$\min_w \|\Phi w - t\|^2.$$

When appropriate, solve $\Phi^T \Phi w = \Phi^T t$; numerically use a stable least-squares solver, especially when columns are dependent.

Positive sample	Score	Residual	Squared loss
Correct but extreme	8	7	49
Wrong but near zero	-.1	-1.1	1.21

The correctly classified extreme point costs much more under squared loss. This can pull the fitted boundary even though its sign is already correct. A classification-specific objective asks a different question; perceptron updates and logistic loss will be developed in later lectures.

Least squares fits a numerical target, not just its sign

For stated classroom points $(x, t) = (-1, -1), (0, 1), (1, 1)$, fit $g = ax + b$ by minimizing

$$E(a, b) = \sum_i (t_i - ax_i - b)^2.$$

The centered inputs have $\sum x_i = 0$, so the normal equations give

$$a = \frac{\sum_i x_i t_i}{\sum_i x_i^2} = \frac{2}{2} = 1, \qquad b = \frac{\sum_i t_i}{3} = \frac{1}{3}.$$

Predicted scores are $(-2/3, 1/3, 4/3)$; all three signs are correct. The residuals are $(-1/3, 2/3, -1/3)$, giving $E = 2/3$.

Criterion distinction

Correct classification does not stop squared loss from penalizing a positive score above its target. The objective cares about numerical target agreement, whereas zero-one classification only checks the final label.

A distant same-class point can move a squared-loss boundary

Add $(x, t) = (10, 1)$ to the preceding three observations. All targets remain ± 1.

$$\bar{x} = 2.5, \quad \bar{t} = .5, \quad \sum x_i^2 = 102, \quad \sum x_i t_i = 12.$$

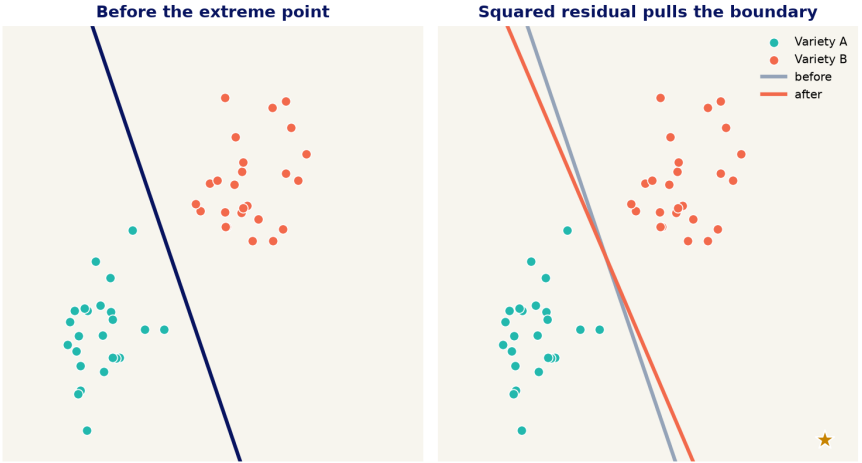
$$a = \frac{12 - 4(2.5)(.5)}{102 - 4(2.5)^2} = \frac{7}{77} = \frac{1}{11}, \qquad b = .5 - \frac{2.5}{11} = \frac{3}{11}.$$

Now $g(-1) = 2/11 > 0$, misclassifying the original negative point. The previous rule $g = x + 1/3$ still assigns every point the correct sign, but predicts a very large score at $x = 10$.

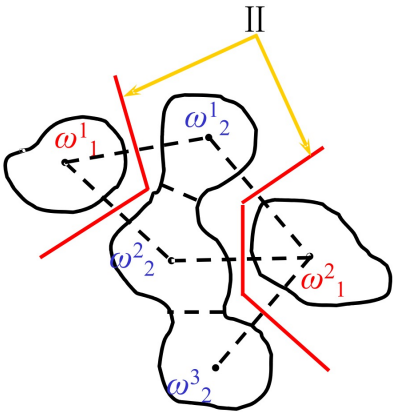
Why the change occurs

Squared loss tries to bring that large positive score back toward target 1. The distant observation has substantial leverage; a label-correct affine rule need not minimize this numerical loss.

A distant same-class point can move a squared-loss boundary



Several prototypes preserve separate class regions



Read the class regions. Red labels mark two subregions of class 1; blue labels mark three of class 2. A point inside each region represents a prototype. The red separating boundary bends

because different nearby prototype pairs supply different local comparisons. Dashed links show center relations, not class boundaries. One mean

for all of class 1 would lie between its separated modes and lose structure. Several prototypes retain local information while keeping each pairwise comparison simple. Prototype count controls how much detail the model can express.

Nearest-prototype scores become affine comparisons

$$\hat{j} = \arg\min_j \|x - \mu_j\|^2.$$

Expand one negative squared distance:

$$-\|x - \mu_j\|^2 = -x^T x + 2\mu_j^T x - \|\mu_j\|^2.$$

The term $-x^T x$ is common to every prototype. Therefore compare affine scores

$$h_j(x) = 2\mu_j^T x - \|\mu_j\|^2.$$

Class score with multiple prototypes.

$$g_c(x) = \max_{j:\, label(j)=c} h_j(x), \qquad \hat{y} = \arg\max_c g_c(x).$$

Each pair meets on a straight bisector. The identity of the nearest prototype changes across space, so the final class boundary consists of different straight pieces. A prototype switch inside one class need not be a class boundary.

Shared covariance gives Mahalanobis geometry

The nearest Euclidean prototype rule matches equal-prior Gaussian comparison when the shared covariance is spherical, $\Sigma = \sigma^2 I$. For general shared Σ, compare

$$d_\Sigma^2(x, \mu) = (x - \mu)^T \Sigma^{-1} (x - \mu).$$

With $q = (0, 0)$, $\mu_A = (1, 10)$, $\mu_B = (3, 1)$, and $\Sigma = \mathrm{diag}(1, 100)$:

$$d_E^2 = (101, 10), \qquad d_\Sigma^2 = (2, 9.01).$$

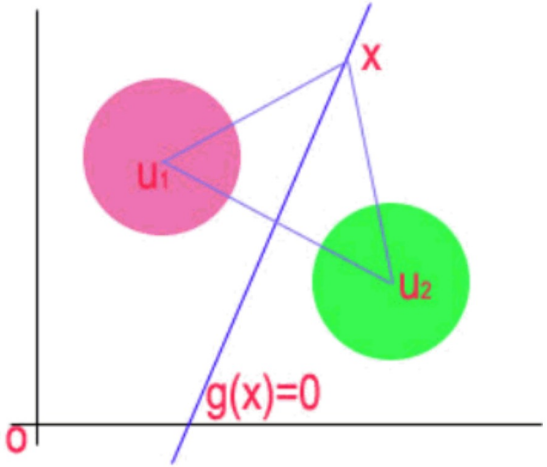
Euclidean comparison selects B; covariance-aware comparison selects A.

The boundary remains affine

Removing the common $x^T \Sigma^{-1} x$ leaves prototype scores $2\mu_j^T \Sigma^{-1} x - \mu_j^T \Sigma^{-1} \mu_j$.

The metric changes which differences matter. Shared covariance alone does not justify the unweighted Euclidean distance used in the simpler prototype example.

Shared covariance gives Mahalanobis geometry



Worked prototype decisions retain a multimodal class

Class A has prototypes $\mu_{A1} = (0, 0)$ and $\mu_{A2} = (4, 0)$; class B has $\mu_B = (2, 2)$.

Input	d^2_{A1}	d^2_{A2}	d^2_B	Prediction
$(.5, .5)$	.5	12.5	4.5	A (prototype 1)
$(3.5, .5)$	12.5	.5	4.5	A (prototype 2)
$(2, 1.5)$	6.25	6.25	.25	B

Using one A mean $(2, 0)$ would discard the two separated A modes. Two prototypes keep both. **Local boundary.** A1 versus B has equation $x_1 + x_2 = 2$; A2 versus B has $-x_1 + x_2 = -2$. Their active segments differ because the nearest A prototype changes.

One mean can fall between two supported class modes

Keep A prototypes $(0, 0)$ and $(4, 0)$ and B prototype $(2, 2)$. Compare $x = (3.8, 1.1)$ under two A representations.

A representation	Squared distance	Comparison with B
One mean $(2, 0)$	$1.8^2 + 1.1^2 = 4.45$	B distance 4.05 wins
Two prototypes	$\min(15.65, 1.25) = 1.25$	A wins

The global A mean summarizes location but discards the two separate concentrations. It need not lie in a densely populated part of A.

Supported complexity

Two prototypes express the stated two-mode geometry. Adding representatives is useful when repeated local observations support them; a prototype placed merely to repair one noisy training point may create an unstable region.

Only active prototype bisectors become class boundaries

The affine prototype scores are $h_{A1} = 0$, $h_{A2} = 8x_1 - 16$, and $h_B = 4x_1 + 4x_2 - 8$. Class B wins when it exceeds both A pieces:

$$x_1 + x_2 \geq 2, \qquad -x_1 + x_2 \geq -2, \quad \text{equivalently } x_2 \geq |x_1 - 2|.$$

The visible A1/B boundary uses $x_1 + x_2 = 2$ only for $x_1 \leq 2$. The A2/B boundary uses $x_2 = x_1 - 2$ only for $x_1 \geq 2$.

Internal switch

A1 and A2 tie on $x_1 = 2$. At $(2, -1)$ A wins on both sides; at $(2, 1)$ B exceeds both. Neither point marks an A/B boundary merely because the A prototype identity switches.

k-nearest neighbors use training samples rather than fixed prototypes

For a new x, find its k nearest labelled training samples and predict the majority class.

Training point; query $x = (2, 1)$	Label	Squared distance
$(2, 1.1)$	A	.01
$(1.5, 1.8)$	B	.89
$(2.5, 1.8)$	B	.89
$(0, 0)$	A	5
$(4, 0)$	A	5

Compare. $k = 1$ predicts A, $k = 3$ predicts B, $k = 5$ predicts A. One possibly noisy nearby label dominates $k = 1$; very large neighborhoods can lose local class structure. Choose k using validation data. Define distance ties and vote ties. Nearest-prototype classification stores representatives; kNN keeps the labelled observations as local evidence.

Distance-weighted neighbors can reverse an unweighted vote

As a teaching extension of local sample evidence, use the earlier query $(2, 1)$ and its three nearest labels: A at distance .1, and two B observations at distance $\sqrt{.89}$.

$$\text{Uniform votes: } A = 1,\ B = 2 \quad \Rightarrow B.$$

With weights $\omega_i = 1/d_i$,

$$\text{Weighted votes: } A = 10, \quad B = 2/\sqrt{.89} \approx 2.12 \quad \Rightarrow A.$$

Different evidence policy

Uniform voting counts observations equally. Distance weighting gives nearby observations more influence, which can also amplify a nearby mislabeled point.

When a query exactly matches a training location, handle zero distance explicitly, for example by voting among those exact matches before applying inverse-distance weights to other observations.

Neighbor identity depends on scaling; output depends on the task

Query $q = (0, 0)$ and candidate neighbors $a = (1, 10)$, $b = (3, 1)$:

Coordinates used	$\|q - a\|^2$	$\|q - b\|^2$	Nearest
Raw features	101	10	b
Second feature divided by 10	2	9.01	a

Fit appropriate scaling from training data; distance units determine which observations are local.

Task	Neighbor output	Example for three neighbors
Classification	Vote or weighted class vote	$(A, B, B) \to B$
Regression	Average or weighted numeric target	$(8, 9, 13) \to 10$

Neighbors define the evidence set; the task defines how their outputs are combined. Neither geometric closeness nor a vote fraction automatically supplies a well-calibrated class probability.

Validation chooses neighborhood size using unseen labels

Consider the following stated classroom validation results on the same ten held-out observations. The distance and preprocessing are fixed.

Neighborhood	Validation errors	Error fraction
$k = 1$	$3/10$	.30
$k = 3$	$1/10$	.10
$k = 5$	$2/10$	.20

Choose $k = 3$ among these candidates, then assess the selected pipeline on a separate test set.

Self-neighbor trap

A 1NN training query that includes itself usually has zero distance and reproduces its own label. This is not independent evidence of future accuracy.

The small validation sample gives a tentative choice, not a precise universal error estimate. Fit scaling on training data and apply it unchanged during validation.

Local fitting focuses on where opposite classes meet

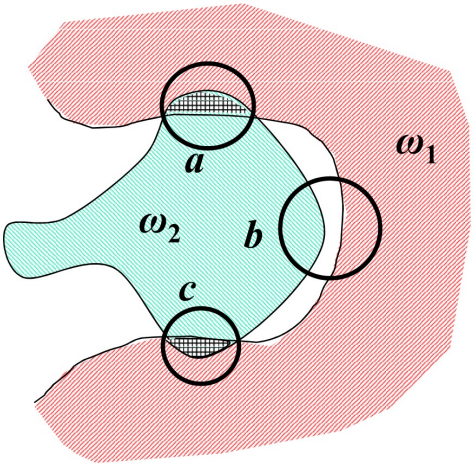
Local-training idea from the Chinese chapter

1. Group each class into supported subregions and select representative prototypes.
2. Find nearby opposite-class prototype pairs, especially mutual nearest pairs.
3. Use their perpendicular bisector as an initial local plane.
4. Refine the local decision using samples near that overlap region.
5. Combine local rules with an explicit class-score or region-assignment policy.

$$\left[x - \frac{\mu_a + \mu_b}{2}\right]^T (\mu_a - \mu_b) = 0.$$

This bisector is an initialization from distance geometry, not proof of an optimal Bayes boundary. Local class shapes, sample support, priors, and losses can require a different final plane.

Local fitting focuses on where opposite classes meet



Mutual nearest prototypes require checking both directions

Use one-dimensional A prototypes $0, 5, 10$ and B prototypes $1, 7, 14$. Their cross-class distances are

	B: 1	B: 7	B: 14
A: 0	1	7	14
A: 5	4	2	9
A: 10	9	3	4

Row minima select $(0, 1)$, $(5, 7)$, and $(10, 7)$. Column minima select $(0, 1)$, $(5, 7)$, and $(10, 14)$. Their intersection contains only $(0, 1)$ and $(5, 7)$.

Local-training role

The selected pairs nominate close opposing regions for local boundary construction. A one-way nearest relation is insufficient; mutuality is a selection heuristic, rather than a proof that every relevant overlap region has been found.

Local planes create a sign code before region-level decisions

Let two local planes give bits $z_1 = \mathbb{I}(x_1 > 0)$ and $z_2 = \mathbb{I}(x_2 > 0)$. Each query first receives a region code, then uses that region's labelled counts.

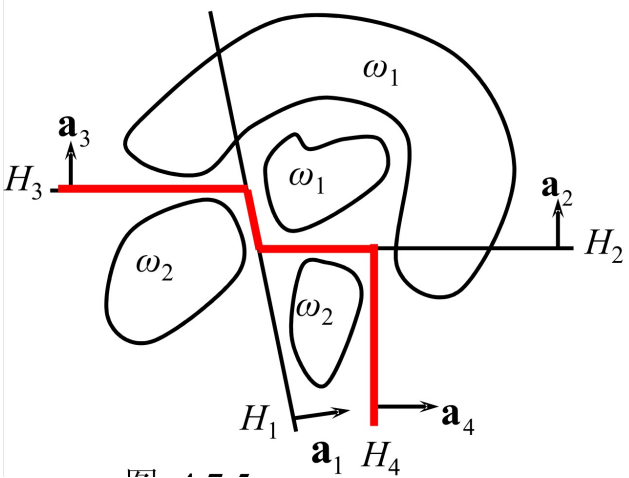
Code	Class-1 count	Class-2 count	Class-1 fraction	Reading
$(0,0)$	1	9	.10	Class 2
$(0,1)$	8	2	.80	Class 1
$(1,0)$	2	8	.20	Class 2
$(1,1)$	1	1	.50	Sparse tie

For $x = (-2, 3)$, the code is $(0,1)$ and the empirical majority selects class 1.

Evidence boundary

Two planes allow at most four codes. Sparse or unseen codes need a stated fallback; a within-region vote fraction is not automatically a calibrated posterior. More planes can increase unsupported cells rapidly.

Local planes create a sign code before region-level decisions



A maximum of affine pieces supplies a definite score

For piece ℓ of class c,

$$g_{c\ell}(x) = w_{c\ell}^T x + b_{c\ell}, \qquad g_c(x) = \max_{\ell} g_{c\ell}(x).$$

$$\hat{y} = \arg\max_{c} g_c(x).$$

A one-dimensional example. Let $g_A(x) = \max(x-3, -x-1)$ and $g_B(x) = 0$.

x	g_A	g_B	Class
-2	1	0	A
0	-1	0	B
4	1	0	A

Class A occupies two tails, while B occupies $-1 < x < 3$. The active A piece changes at $x = 1$, but this is not a class boundary. The class boundaries occur at $x = -1$ and $x = 3$.

More pieces need more supported local evidence

Model size	What it can express	Risk to inspect
One affine score	One broad half-space comparison	Repeated systematic errors on curved structure
A few supported pieces	Several stable local modes	Sparse transition regions
Many tiny pieces	Detailed training gaps	Sensitivity to noise or mislabeled points

Controlled response. Add a prototype or feature when it explains a repeated class pattern. Compare validation predictions before and after that change. Keep enough observations supporting each local region. The simplest useful model is defined by the data geometry and learning objective. A complicated boundary that tracks each training sample can fail between observations even while its training error is zero.

A model comparison must include validation error

Suppose a stated classroom comparison uses 40 training observations and the same 20 validation observations.

Representation	Training errors	Validation errors
One affine score	$6/40 = .15$	$4/20 = .20$
Selected quadratic features	$2/40 = .05$	$2/20 = .10$
Many small prototype regions	$0/40 = 0$	$5/20 = .25$

Among these candidates, quadratic features provide the strongest validation result despite imperfect training labels. The many-prototype model fits every training label but does worse on unseen observations.

Interpret before choosing

Inspect which geometric mistakes persist and whether each added feature or local region has enough support. The table supports a provisional choice; an independent test set is needed after model selection, and these small counts leave uncertainty about future performance.

Integrated exercise: choose and compute the representation

1. **Interval.** Class B for $2 < x < 5$, class A outside. Write one quadratic score and its feature weights.
2. **Diagonal corners.** Labels are positive for equal-sign corners of $(\pm 1, \pm 1)$. Give a one-feature decision rule and explain why an affine score fails.
3. **Multimodal class.** A prototypes $(0, 0), (4, 0)$; B prototype $(2, 2)$. Classify $(3, 1)$ by squared distances. State a tie rule if needed.

Explain your choice. Match the representation to the repeated geometry, then state how its score creates the label. More parameters are justified only when they express a needed pattern that simpler features miss.

Solutions: simple calculations, different geometry

Problem	Computation	Rule
Interval	$(x-2)(x-5) = x^2 - 7x + 10$	Negative: B; positive: A
Diagonal corners	$g = x_1 x_2$	Sign gives the target
Prototype point $(3, 1)$	Distances squared $(10, 2, 2)$	A2/B tie; use declared tie policy

For the interval, $\phi = (x^2, x, 1)$ and $w = (1, -7, 10)$. For the corner pattern, adding an interaction changes the representation, so the old affine impossibility proof no longer applies. For the prototype tie, the distance model has no preference; a review or fixed class-order rule must be added explicitly. Probabilities and costs would supply another decision layer only if a suitable model is provided.

What to explain before choosing a learning algorithm

Layer	Question	This lecture's answer
Representation	What information enters the score?	Raw, mapped, or local features
Geometry	Which regions can the score express?	Hyperplane, conic, or piecewise boundary
Learning criterion	What makes a direction or fit good?	Fisher ratio, target fit, supported local evidence
Decision	How does the output become a label/action?	Sign, argmax, threshold, and tie/loss rule

Exit explanation. Describe w as a boundary normal, explain both parts of $J(w)$, and give a dataset where a feature map is necessary. Next lecture learns a separating score by correcting mistakes and builds decision trees by asking successive questions. The geometry and objective determine what those training rules can achieve.